Add description to FANUC Robot series R-30*i*B/R-30*i*B Mate CONTROLLER OPERATOR'S MANUAL (Collaborative Robot Function)

1.Type of applied technical documents

Name	FANUC Robot series R-30*i*B/R-30*i*B Mate CONTROLLER OPERATOR'S MANUAL (Collaborative Robot Function)
Spec.No./Ed.	B-83744EN/01

2.Summary of Change

Group	Name/Outline	New, Add, Correct, Delete	Applicable Date
Basic	Following descriptions are added. - Addition of screen items (Section 2) - Payload Confirmation without Teach Pendant (Section 3) - Collaborative Robot Monitor (Section 4)	Correct Add	Immediately
Optional Function			
Unit			
Maintenance Parts			
Notice			
Correction			
Another			

Change of Section 2. Underlined part is the change point.

2.2 SETTING COLLABORATIVE ROBOT FUNCTION

This section introduces the setting of the collaborative robot function. In the collaborative robot screen, you can set up the collaborative robot function.

Collaborative robot screen

1. Press the [MENU] key.
2. Select SYSTEM.
3. Press F1, [TYPE].
4. Select DCS.
5. If neither DCS top screen nor Collaborative robot screen, press the [PREV] key, and select Collaborative robot. The following screen will be displayed. Some items is not displayed depending on the software version.

```
DCS
    Collaborative robot

    Contact stop status   STOP
    Enable/Disable        ENABLE         OK
    Force sensor
       Serial number:           0        OK
    Group:                      1        OK
    Payload setup:        <DETAIL>       OK
    Active Payload number: No. 1 [**********]
    External force Limit / Disabling input
      Current:      0.0
      Limit 1:   150.00[N] ---[  0]       OK
      Limit 2:     0.00[N] ---[  0]       OK
      Limit 3:     0.00[N] ---[  0]       OK
      Limit 4:     0.00[N] ---[  0]       OK
      Escape:    300.00[N]                OK
    Force Monitor:   Current   Peak
              0.00[N] (    0%)    0%
     Settings  :      Warning   Output
                         0%    GO[  0]
    Payload Monitor: Current   Peak
      Force    :           0%     0%
      Moment   :           0%     0%
      Settings  :     Warning   Output
         Force  :          50%    GO[  0]
         Moment :          50%    GO[  0]
         Warning in DSBL:          DISABLE
    Payload change distance (mm)
           Current   +Limit   -Limit
      X:      0.00    0.00    0.00      OK
      Y:      0.00    0.00    0.00      OK
      Z:      0.00    0.00    0.00      OK
      Enabling input:     ON [  0]       OK
    Password for CONFIRM:   ENABLE       OK
    CONFIRM by DI:          DISABLE      OK
      CONFIRM input:        DI[  0]
      Payload No. output:   GO[  0]
    Speed limit:  250.00 [mm/s]          OK

   [TYPE]  CONFIRM  PEAKCLR         UNDO
```

You can set up the collaborative function in this screen.
For configurable items, OK/CHGD/PEND is displayed in the right.
OK: Setting parameter and DCS parameter are the same.
CHGD: Setting parameter is changed, but not applied to DCS parameter.
PEND: Setting parameter is changed and applied to DCS parameter, but controller power has not been cycled.

When you change DCS parameter setting, "Apply to the DCS parameter" is necessary. Refer to "Dual Check Safety Function OPERATOR'S MANUAL"(B-83184EN), Chapter of OVERVIEW, Section of APPLY TO DCS PARAMETER.

Contact stop status

The status of the contact stop function is displayed. Refer to Section 2.3.

Enable/Disable (configurable)

Enabling or disabling the contact stop function.

DISABLE

The contact stop function is disabled. In this setting, the robot is same as a standard robot. You can move the robot only in T1 mode. In T2 or AUTO mode, "SYST-343 Enable Contact Stop function" occurs, and you cannot move the robot.

ENABLE

The contact stop function is enabled.

ENBL(Shift+Reset)

Contact stop is always enabled in AUTO mode. When you hold down the [SHIFT] key and press the [RESET] key in T1 or T2 mode and the contact stop status is STOP (Refer to Section 2.3), the contact stop function is disabled until the [SHIFT] key is released or the safety is confirmed.

Refer to Subsection 4.1.1 for the detail of the difference between ENABLE and ENBL(Shift+Reset).

> ⚠**CAUTION**
> When "ENBL(Shift+Reset)" is selected, the contact stop function might be disabled when you hold down the [SHIFT] key and press the [RESET] key in T1 or T2 mode. If "ENBL(Shift+Reset)" is used, adequate risk assessment for the whole the robot system is necessary to verify that the contact stop can be disabled while Shift+Reset is pressed in T1 or T2 mode.

Force Sensor Serial number

The serial number of the force sensor is displayed.

Group (configurable)

Set the group number of the collaborative robot.

Payload setup

You can see current payload setup. Refer to Subsection 2.2.1.

Active payload number

Current payload number is displayed and not configurable in this screen. You can select the payload number in "SYSTEM > Motion" screen or by PAYLOAD instruction in program. Refer to "R-30*i*B/R-30*i*B Mate CONTROLLER OPERATOR'S MANUAL (Basic Operation)"(B-83284EN) .

External force (N)

Current

You can see the current external force. This item moved to Force monitor in the software version 7DC3/30 or later.

Limit 1 to 4 (configurable)

Set the external force limit 1 to 4 between 0[N] and 150[N], inclusive. 0[N] means disable and you cannot set the limit 1 to 0[N].
When the contact stop function is enabled, "@" is displayed as following.

```
 Limit 1:   150.00[N] ---[  0]       OK
@Limit 2:    90.00[N] ---[  0]       OK
 Limit 3:     0.00[N] ---[  0]       OK
 Limit 4:     0.00[N] ---[  0]       OK
```

This means the active force limit. The active force limit is minimum limit in available limits. When the external force exceeds the active force limit, the robot will stop. To disable the force limit, set the limit to 0[N] or the corresponding disabling input to ON. You can disable the force limit to change the active force limit dynamically in a program.

In the example of this screen, the limit 1 and 2 are enabled, then @ is attached to the minimum limit 2. The limit 3 and 4 is set 0[N], then they are disabled.

Escape (configurable)

Set the external force limit for Push To Escape between 0[N] and 300[N], inclusive. In Push To Escape, if the external force exceeds this limit, the robot is stopped. This means that the robot does not escape when you push with a force more than the force limit.

Disabling input (configurable)

Set the Safe I/O to disable each force limit. When the disabling input is ON, the corresponding force limit is disabled. When the all limits are disabled, the contact stop function is disabled.

```
@Limit 1:   150.00[N] SIR[  1]       OK
 Limit 2:    90.00[N] SIR[  2]       OK
 Limit 3:     0.00[N] ---[  0]       OK
 Limit 4:     0.00[N] ---[  0]       OK
```

In the example of this screen, SIR[2] is ON to disable the limit 2. The limit 1 is only enabled, then @ is attached to the limit 1.

Force Monitor/Payload Monitor

These items is displayed in the software version 7DC3/30 or later. An external force and a load applied to the robot is displayed. You can configure the warning posted when the monitored value exceeds specified value, and output monitored value. Refer to Subsection 4.2.3.

Payload change distance

Current

You can see the current distance and the rotation angle of the posture from a payload change position. The rotation angle of the posture is displayed in the software version 7DC3/30 or later. When the payload changing output is OFF (Refer to Section 2.3), this value is 0.

+Limit, -Limit, Rotation Limit (configurable)

When the payload change distance is enabled, the active payload number is changed, then the contact stop function will switch to disable. If the face plate position is changed more than the payload change distance, the contact stop function switches to enable immediately. The payload change distance can be set for each direction individually, +X, -X, +Y, -Y, +Z and –Z of the world coordinate system. Rotation Limit is displayed in the software version 7DC3/30 or later. Rotation Limit is one value without distinction of direction. This is useful for the collaborative robot to pick or place a workpiece. Refer to Chapter 5.

Enabling input (configurable)

Set the Safe I/O to enable the payload change distance. When the enabling input is ON, the payload change distance is enabled. When the Safe I/O is OFF, the payload change distance is disabled, and STOP state (Refer to Section 2.3) will be initiated immediately when the actual payload number is changed. Default setting is ON[0]. It means the payload change distance is always enabled.

Password for CONFIRM (configurable)

After power on the controller, the payload confirmation operation must be performed to clear the alarm. The collaborative robot does not move until the payload confirmation is performed. When Password for CONFIRM is ENABLE, DCS password is required for the payload confirmation operation. Default setting is ENABLE. Refer to Chapter 3.

CONFIRM by DI (configurable)

This item is displayed in the software version 7DC3/30 or later. You can perform the payload confirmation operation with some kind of terminal without other than the teach pendant. Refer to Chapter 3.

Speed limit (configurable)

This item is displayed in the software version 7DC3/28 or later. When the contact stop is enable, the speed of the collaborative robot exceeds the speed limit, “SYST-323 Collaborative speed limit” will be posted, and the robot will stop.

> **⚠WARNING**
>
> If the speed limit is set to inappropriate value, the force at the time of contact increases, and a serious personal injury could result. Risk assessment for the whole robot system is required to determine appropriate speed limit. Refer to followings about the risk according to the speed.
>
> CR-35*i*A: MECHANICAL UNIT OPERATOR’S MANUAL(B-83734EN), Section 3.6 THE CHARACTERISTIC OF COLLABORATIVE ROBOT AND LIMITATIONS AND USAGE NOTES
>
> CR-4*i*A, CR-7*i*A, CR-7*i*A/L: MECHANICAL UNIT OPERATOR’S MANUAL(B-83774EN), Section 3.6 THE CHARACTERISTIC OF COLLABORATIVE ROBOT AND LIMITATIONS AND USAGE NOTES

F2 [CONFIRM]

Pressing the F2 [CONFIRM] and the payload confirmation starts. The payload confirmation operation must be performed at least once after power on. The collaborative robot does not move until the payload confirmation is performed. Refer to Chapter 3.

⚠WARNING

1. If the collaborative robot settings are incorrectly, the safety function will not work, and a serious personal injury could result. When the collaborative robot settings are changed, the values must be verified and the function must be tested again.
2. When the contact stop is disabled, the collaborative robot does not stop even though the external force exceeds the limit, and a serious personal injury could result. If the robot system is designed to be able to disable the contact stop, adequate risk assessment for the whole robot system is necessary to verify that the contact stop can be disabled.
3. If “Password for CONFIRM” is set to DISABLE, anybody can perform the payload confirmation operation. If the payload confirmation operation is performed incorrectly, the external force is not detected correctly and safety function will not work, and a personal injury could result. When “Password for CONFIRM” is set to DISABLE, everybody who can operate the teach pendant must be authorized to perform the payload confirmation operation.

2.2.1 Verify Payload Setup

In the collaborative robot screen, select Payload setup: <DETAIL>, then the payload setup screen is displayed. In this screen, current parameters for the contact stop are displayed as followings. In this screen, you cannot change their parameters.

```
DCS
    Payload setup
                                          Status
    PAYLOAD  1
      PAYLOAD    [kg]            35.00    OK
      CENTER X   [cm]             0.00    OK
      CENTER Y   [cm]             0.00    OK
      CENTER Z   [cm]             0.00    OK
      INERTIA X  [kg cm^2]        0.00    OK
      INERTIA Y  [kg cm^2]        0.00    OK
      INERTIA Z  [kg cm^2]        0.00    OK
    PAYLOAD  2
      PAYLOAD    [kg]            35.00    OK

  [TYPE]
```

PAYLOAD 1 to 9

Payload settings are displayed. In “SYSTEM > Motion” screen, you can change the payload settings.

```
DCS
    Payload setup
                                        Status
    ARM LOAD #3
      PAYLOAD    [kg]             0.00     OK
      CENTER X   [cm]            -3.50     OK
      CENTER Y   [cm]             0.00     OK
      CENTER Z   [cm]             8.00     OK
    ARM LOAD  #0
      PAYLOAD    [kg]             0.00     OK
      CENTER X   [cm]             0.00     OK
      CENTER Y   [cm]             0.00     OK
      CENTER Z   [cm]             0.00     OK
    ARM LOAD  #0

  [TYPE]
```

ARM LOAD #axis_number

Arm load settings are displayed. In “SYSTEM > Motion” screen, you can change the payload settings.

```
DCS
    Payload setup
                                        Status
    Gravity vector
        X [mm/s^2]               0.00     OK
        Y [mm/s^2]               0.00     OK
        Z [mm/s^2]            9800.00     OK
    Speed limit for push to escape
        J1 [deg/s]               9.00     OK
        J2 [deg/s]               9.00     OK
    Monitor interval(ms):         500     OK
    Vibration limit(N):         30.00     OK
    Payload error limit(N):     60.00     OK
    Status chk err lim(N):      60.00     OK
    Status chk time lim(s):     90.00     OK
    Vibration tolerance:           32     OK
    Payload change distance
     Rotation limit (deg):       0.20     OK

  [TYPE]
```

Gravity vector

Gravity vector is displayed. The gravity vector is changed by the mount angle of the robot.

Speed limit for push to escape

Speed limit in Push To Escape is displayed.

Monitor interval

Sampling time of the external force monitor is displayed. Refer to the section 4.2.

Vibration limit

The force limit by vibration is displayed.

Payload error limit

Payload error limit is displayed. When the error of the payload setting exceeds the payload error limit, the robot can not move. Refer to Subsection 4.2.2.

Status chk error lim

This is “Status check error limit”, internal parameter for the auto status check.

Status chk time lim

This is “Status check time limit”, the limit of time interval for the auto status check.

Vibration tolerance

Tolerance of floor vibration

Payload change distance Rotation limit (deg)

The orient rotation limit of the payload change distance. <u>This item is displayed in the software version before 7DC3/29.</u>

> ⚠**WARNING**
> If the payload setting is incorrect, the safety function will not work correctly, and a personal injury could result. When payload settings are changed, the values must be verified and the function must be tested again.

Change Section 3.1 is added.

3.1 Payload Confirmation without Teach Pendant

When you would like to perform the payload confirmation operation with some kind of terminal other than the teach pendant, configure “CONFIRM by DI” in the collaborative robot screen. This config is displayed in the software version 7DC3/30 or later.

```
DCS
    Collaborative robot

    CONFIRM by DI:           DISABLE       OK
      CONFIRM input:         DI[  0]
      Payload No. output:    GO[  0]

 [TYPE]   CONFIRM   PEAKCLR             UNDO
```

CONFIRM by DI (configurable)

When this item is set to ENABLE, you can perform the payload confirmation operation with an external signal. If it is ENABLE, you can also perform the payload confirmation operation with the teach pendant.

CONFIRM input (configurable)

When “CONFIRM by DI” is ENABLE and you change CONFIRM from OFF to ON, the payload confirmation will be performed. When the CONFIRM input changes from OFF to ON, you need to confirm that the actual payload of the robot hand/tool/workpiece is surely equal to current payload setting and nobody contacts with the robot. When CONFIRM input switches over and over in a short time, the payload confirmation may be performed incorrectly. Keep ON state for more than 1 second.

Payload No. output (configurable)

Current payload setting number is output to a group output signal or a register set in this item. When you perform the payload confirmation operation with some kind of terminal other than the teach pendant, you can confirm the actual payload of the robot hand/tool/workpiece is surely equal to current payload setting by display of current payload setting number on the terminal.

> ⚠**WARNING**
>
> If "CONFIRM by DI" is set to ENABLE, you set CONFIRM input to ON, it is necessary to confirm that the actual payload of the robot matches current payload setting and anybody does not contact the robot. Measure is necessary to prevent putting CONFIRM input by unauthorized person such as locking. If the payload confirmation operation is performed incorrectly, the external force is not detected correctly and safety function will not work, and a personal injury could be result.

Change Subsection 4.2.3 is added.

4.2.3 Collaborative Robot Monitor Screen

"Force Monitor" and "Payload Monitor" are displayed in the collaborative robot screen in the software version 7DC3/30 or later. You can monitor an external force and a load applied to the robot.

```
DCS
    Collaborative robot

    Force Monitor:   Current    Peak
               0.00[N] (    0%)    0%
     Settings  :      Warning    Output
                            0%    GO[  0]
    Payload Monitor: Current    Peak
      Force     :          0%      0%
      Moment    :          0%      0%
      Settings  :     Warning    Output
         Force  :          50%    GO[  0]
         Moment :          50%    GO[  0]
         Warning in DSBL:          DISABLE

   [TYPE]  CONFIRM  PEAKCLR           UNDO
```

Force Monitor

External force applied to the robot is displayed and output to external signals. You can monitor instant force such that a human contact.

Current[N]

Current external force applied to the robot is displayed.

Current[%]

Current force / Active Force Limit x 100 (When the contact stop function is disabled, the value is 0%.)
When the value is 100%, a contact stop occurs. (SYST-320 "Contact force exceeds limit" is posted.)

Payload Monitor

Steady load applied to the robot is displayed and output to external signals. When the payload setting does not match the actual payload, this value will be large.

Current Force[%]

Averaged force / tolerance x 100 (During the payload change, the value is 0%.)
When the value is 100%, payload error occurs. (SYST-325 “Payload error is detected” is posted.)

Current Moment[%]

Averaged moment / tolerance x 100 (During the payload change, the value is 0%.)
When the value is 100%, payload error occurs. (SYST-325 “Payload error is detected” is posted.)

Peak

Peak hold value of each item is displayed. Press F3[PEAK CLR] to clear the value.

Warning

When the current value[%] exceeds each warning value[%], following warning occurs.

- SYST-350 Force Monitor warning
- SYST-348 Payload Monitor (Force) warning
- SYST-349 Payload Monitor (Moment) warning

Set integer value from 0 to 100. When the value is 0, warning does not occur.

Warning in DSBL

DISABLE: No warning occurs when the contact stop function is disabled.
ENABLE: Warnings also occur when the contact stop function is disabled.

Output

Each items can be output to signals etc. Following signal types can be set.

GO: Current value is output to Group output.
R: Current value is output to Numeric Register.
DO/RO: When the current value exceeds the set warning value, signal is set to ON.